



Statistical Reasoning 2

PLAN FOR TODAY!

- ☐ Recap
- ☐ Examining Distributions Cont.
- ☐ Exercises
- ☐ Examining Relationships
- ☐ Exercises

ANNOUNCEMENT

- The prompt for your semester project is online. Please familiarize yourself with the prompt. We'll discuss your questions in the next class.
 - **Deliverable 1**: March 24th, 11:59 pm
 - **Deliverable 2**: April 5th, 11:59 pm
 - **Deliverable 3**: April 19th, 11:59 pm
 - **Deliverable 4**: May 1st, 11:59 pm

ANNOUNCEMENT

Phase A: Foundation & Question Development

Step 1: Topic Exploration & Initial Question Based on the earlier course readings (Creswell; Barroga), identify a broad area of scientific interest. Formulate an initial, draft quantitative research question. This question should aim to explain a relationship, not just describe a phenomenon.

Step 2: Question Refinement Meet with the instructor during class or office hours to refine your question. The goal is to make it more focused, feasible, and clearly connected to a potential causal or correlational relationship.

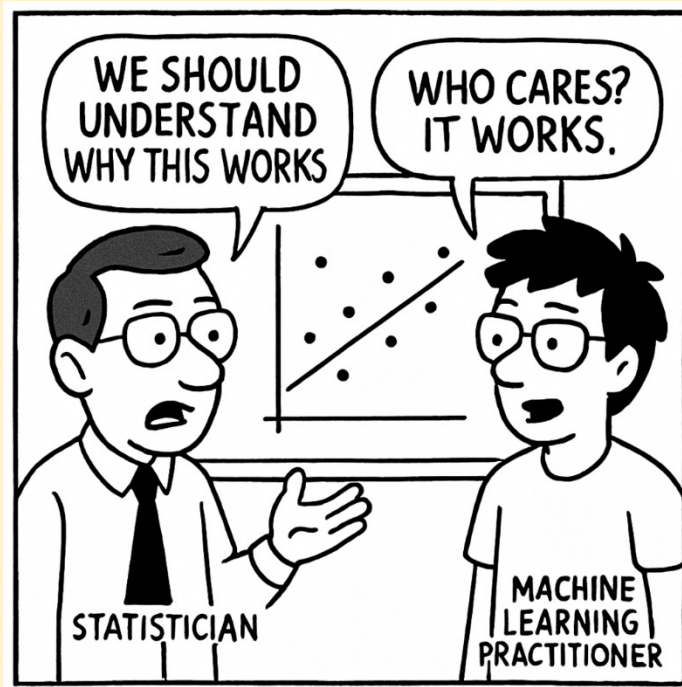
– *Guiding Questions: What is the relationship you are interested in? What are your key concepts?*

RECAP

- 'Statistics' - *converting data into useful information*

The core purpose of statistics is to convert raw data into useful information through the 1) collection, 2) analysis, and 2) interpretation of data.

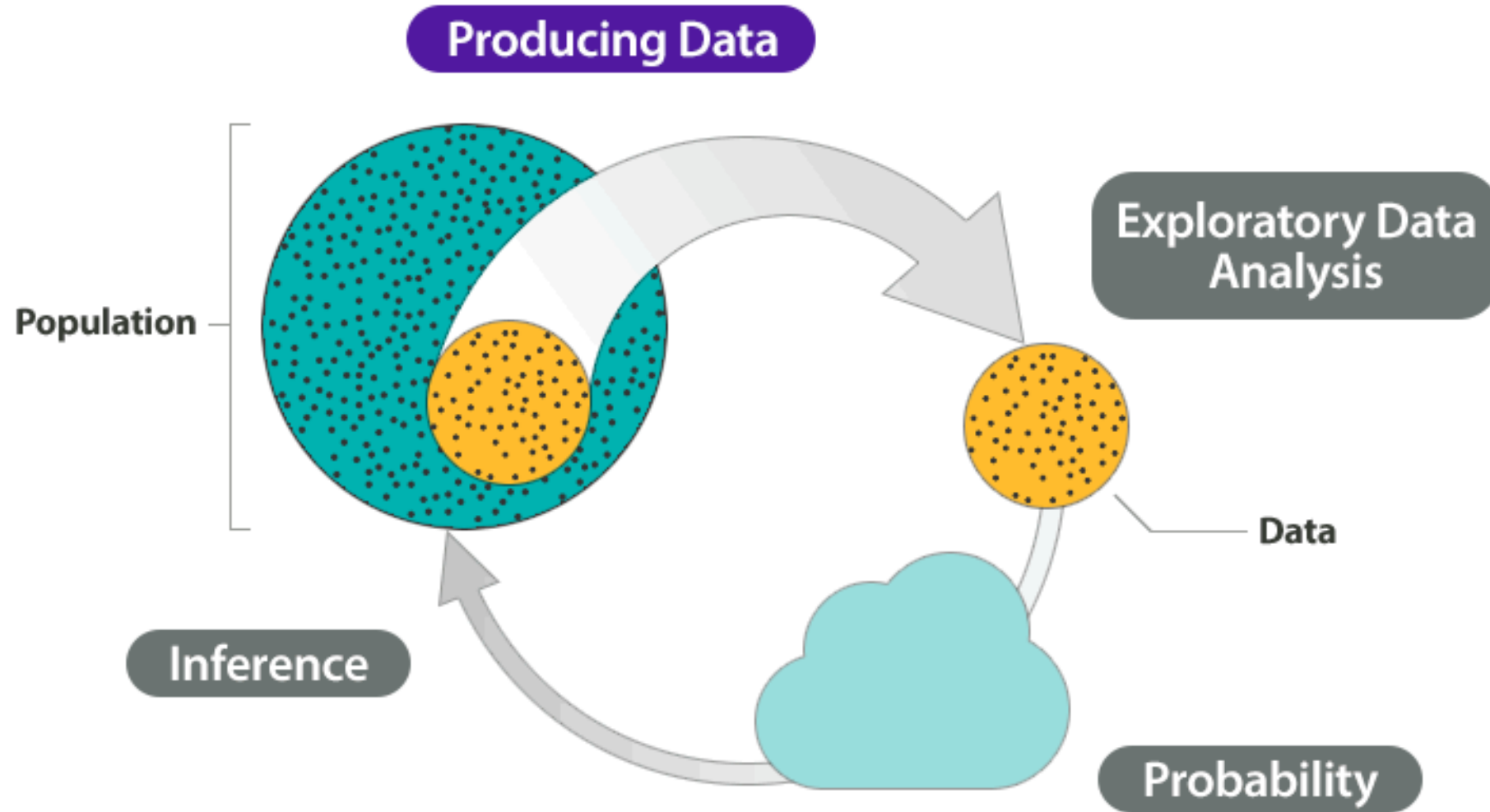
The 'Why' part - allows for drawing conclusions, identifying patterns, and making informed decisions in various fields like science, business, and research.



This is the Big Picture of statistics

Last time!

- 'Statistics' - *converting data into useful information*

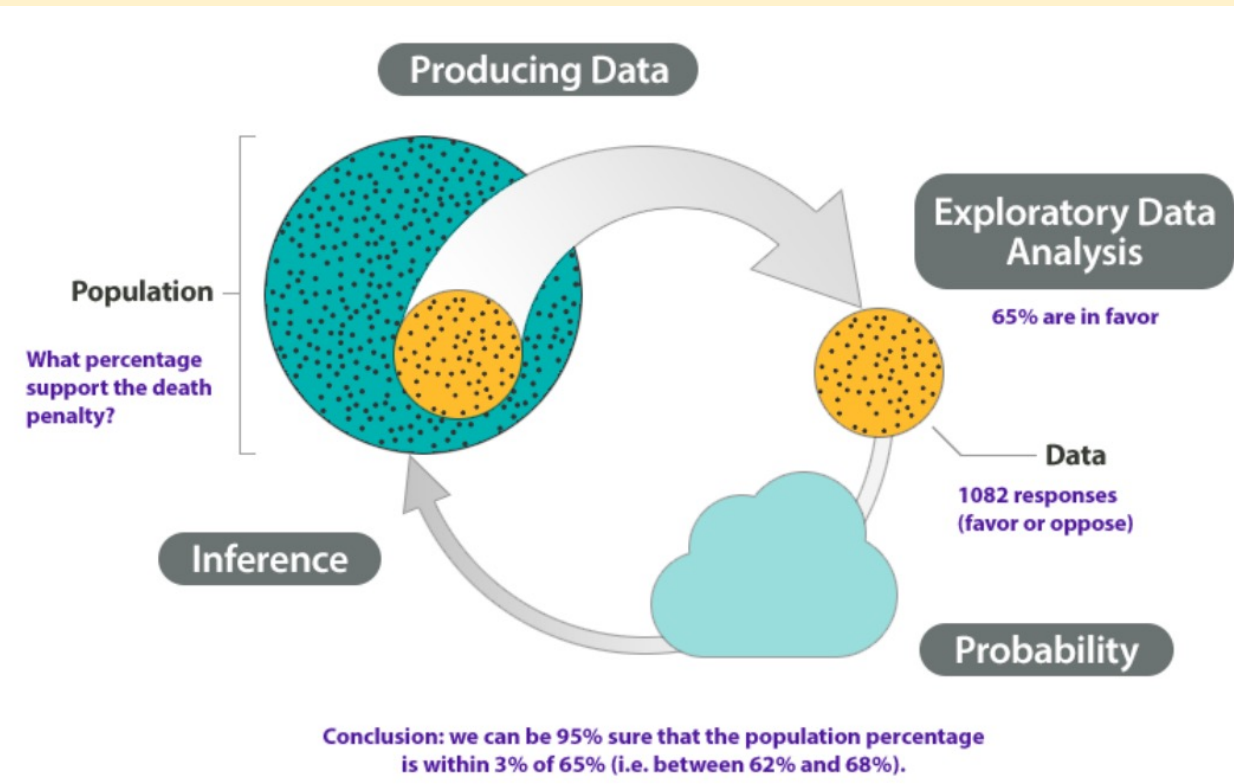


Example

Last time!

At the end of April 2005, a poll was conducted (by ABC News and the Washington Post) for the purpose of learning the opinions of U.S. adults about the death penalty.

1. Producing Data: A (representative) **sample of 1,082 U.S. adults was chosen**, and each adult was asked **whether he or she favored or opposed the death penalty**.
2. Exploratory Data Analysis (EDA): The collected data were summarized, and it was found **that 65% of the sampled adults favor the death penalty for persons convicted of murder**.
- 3 & 4. Probability and Inference: Based on the sample result (of 65% favoring the death penalty) and our knowledge of probability, **it was concluded (with 95% confidence)** that the percentage of those who favor the death penalty in the population is **within 3% of what was obtained in the sample (i.e., between 62% and 68%)**.

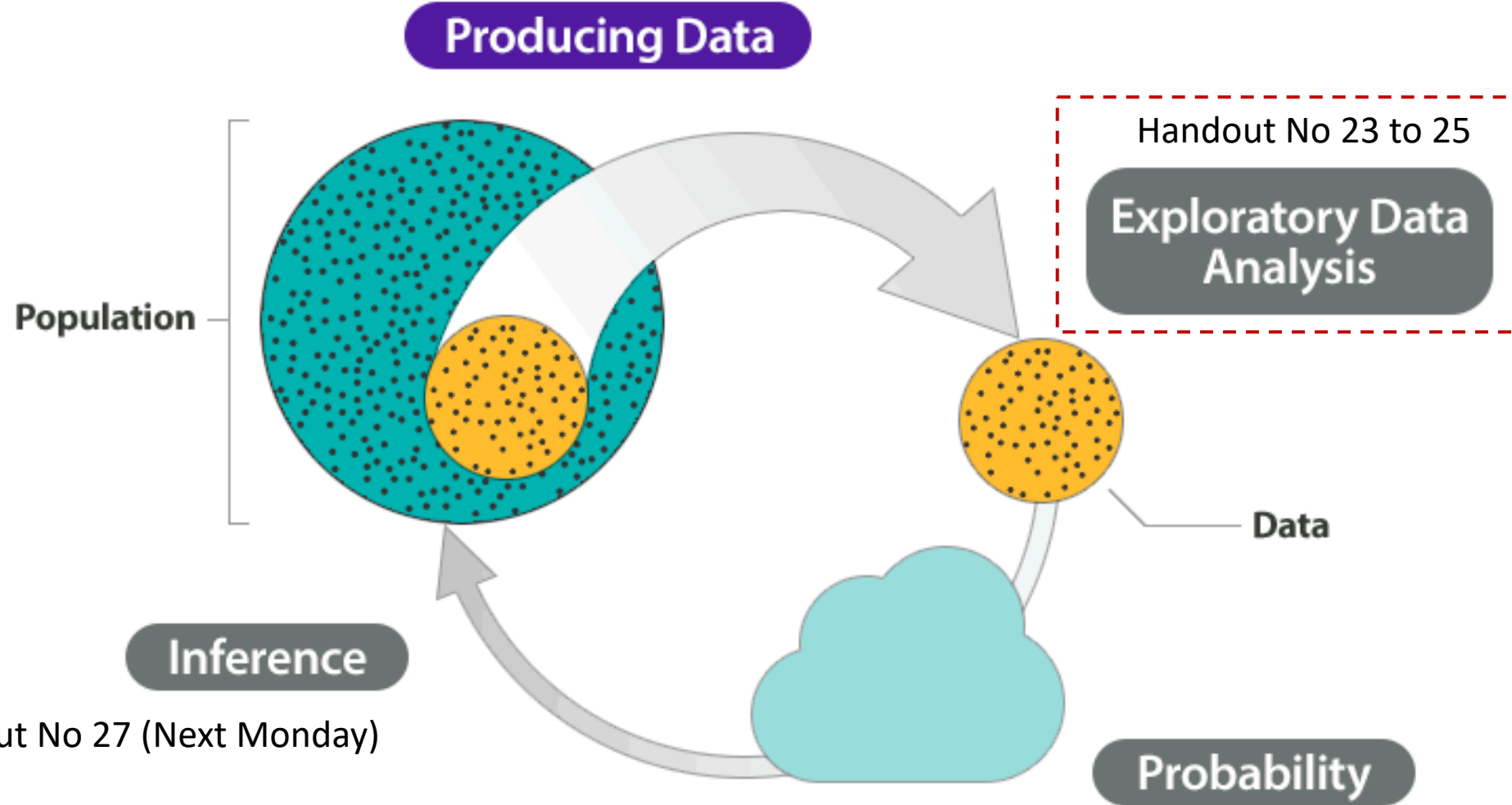


This is the Big Picture of statistics

Last time!

- 'Statistics' - *converting data into useful information*

Handout No 26 (Wednesday)



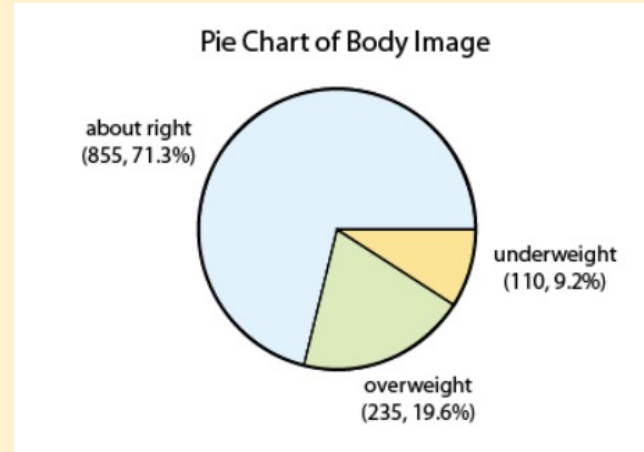
Handout No 27 (Next Monday)

Exploratory Data Analysis

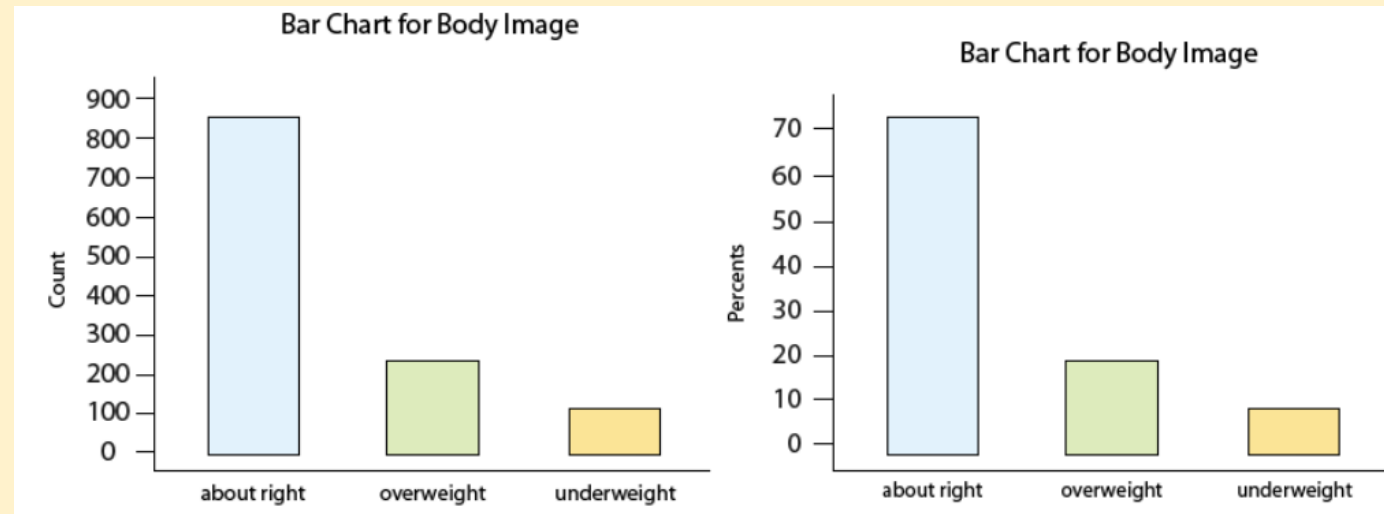
Last time!

- Variables can be classified into one of two types: 'categorical' or 'quantitative'
- One Categorical Variable**

1. The Pie Chart



2. Bar Chart

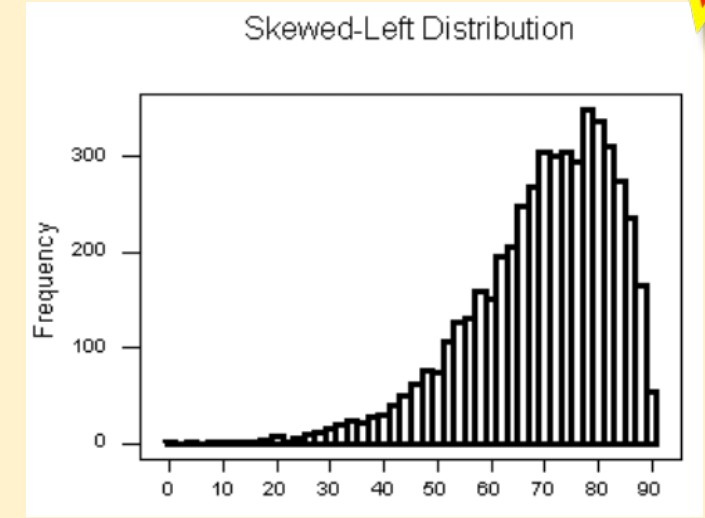
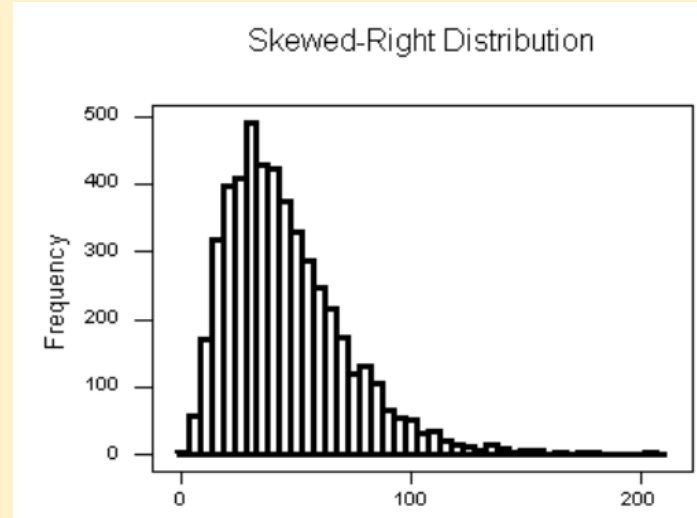
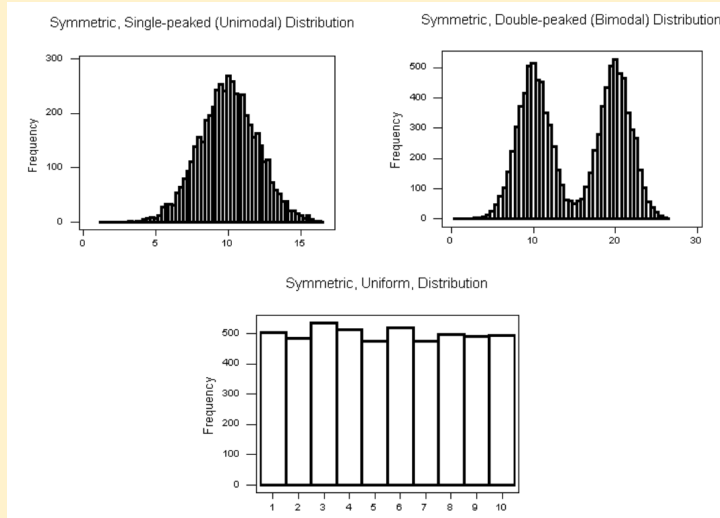


Exploratory Data Analysis

Last time!

One Quantitative Variable

1. Histogram



The overall pattern of the distribution of a quantitative variable is described by its shape, center, and spread

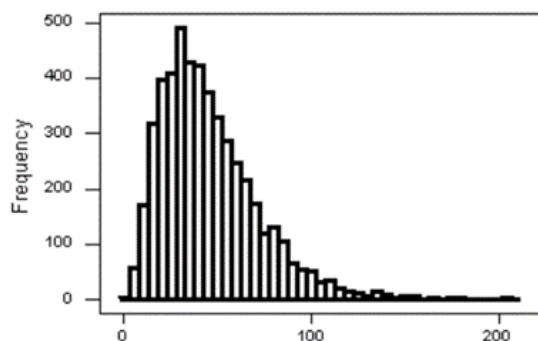
- Shape
 - Center
 - Spread
 - Outliers
- } overall pattern
→ deviations from the pattern

The three main numerical measures for the center of a distribution are the mode, the mean and the median.

Last time!

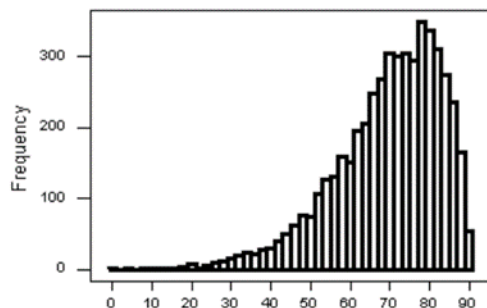
STATISTICS – THE BIG PICTURE

Skewed-Right Distribution



Skewed Right Distributions: the right tail (larger values) is much longer than the left tail (small values)

Skewed-Left Distribution



Skewed Left Distributions: the left tail (smaller values) is much longer than the right tail (larger values).

STATISTICS – THE BIG PICTURE

- **Mean:** is the average of a set of observations (i.e., the sum of the observation divided by the number of observations).

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

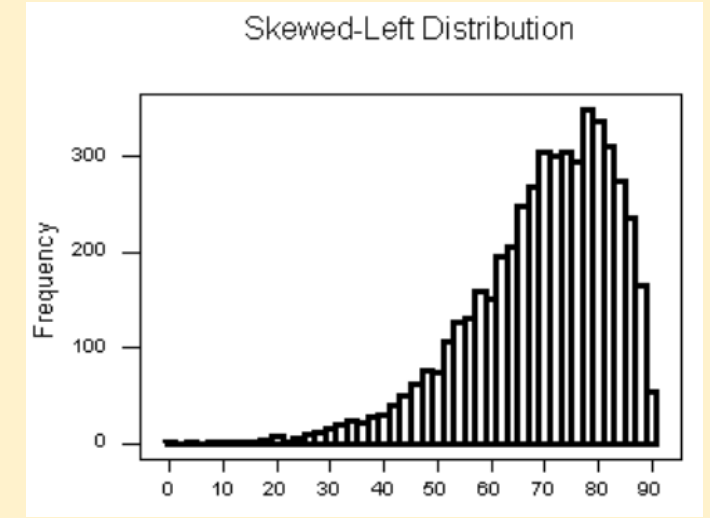
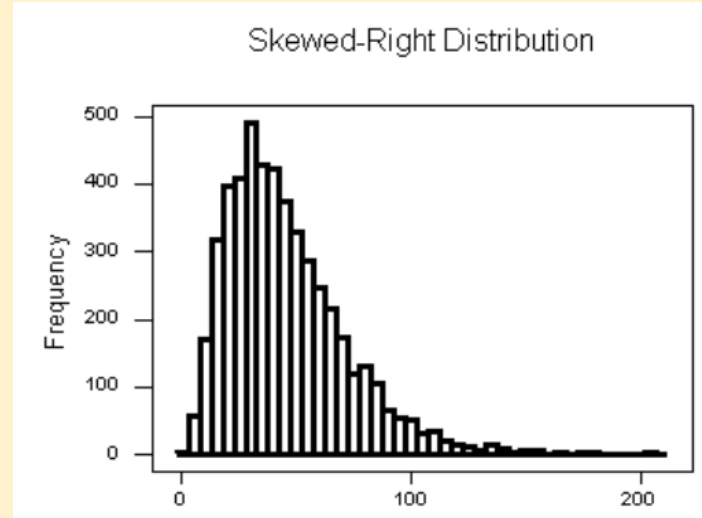
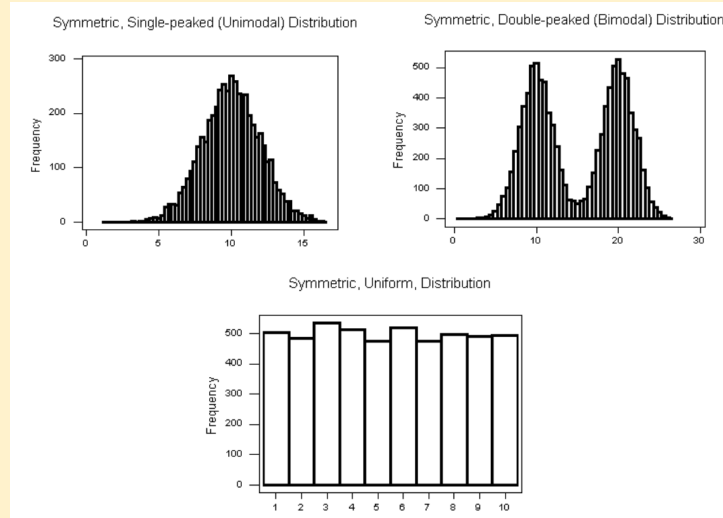
- **Median:** is the midpoint of the distribution. It is the number such that half of the observations fall above, and half fall below.
 - If n is *odd*, the median M is the center observation in the ordered list. This observation is the one “sitting” in the $(n + 1)/2$ spot in the ordered list.
 - If n is *even*, the median M is the *mean* of the *two center observations* in the ordered list. These two observations are the ones “sitting” in the $n/2$ and $n/2 + 1$ spots in the ordered list.
- **Mode:** is the value where the distribution has a “peak”.

EXAMINING DISTRIBUTIONS CONT.

Exploratory Data Analysis

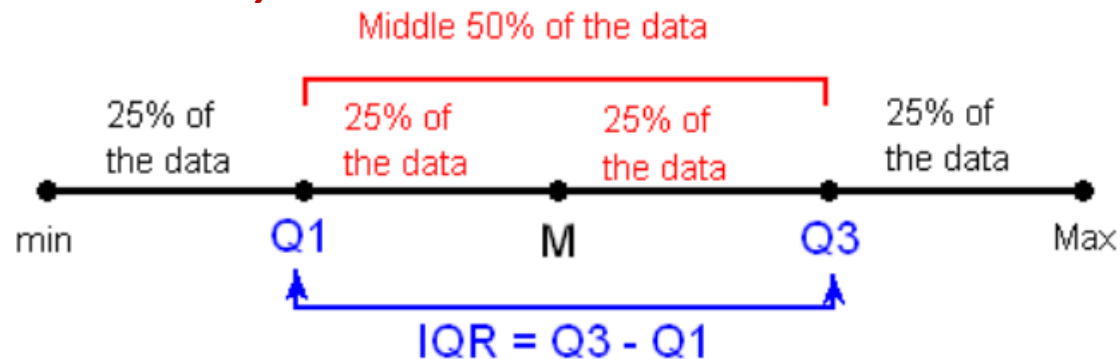
- One Quantitative Variable

- 1. Histogram



- The three most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) Standard deviation

5 number summary



EXAMING DISTRIBUTIONS

- **Range:** The range covered by the data is the most intuitive measure of variability. The range is exactly the distance between the smallest data point (min) and the largest one (Max).
- **Range = Max – min**

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30
29 61 32 33 45 29 62 22 44

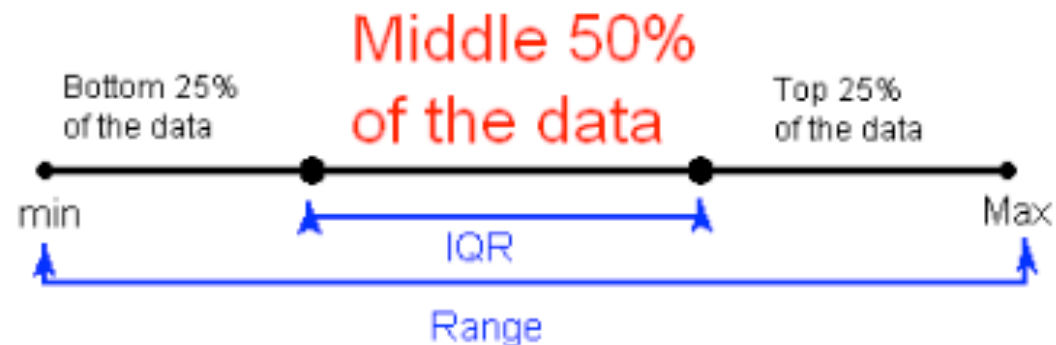
In this example:

- min = 21 (Marlee Matlin for *Children of a Lesser God*, 1986)
- Max = 80 (Jessica Tandy for *Driving Miss Daisy*, 1989)

The range covered by all the data is $80 - 21 = 59$ years.

EXAMING DISTRIBUTIONS

- **Inter-quartile range (IQR)**: While the range quantifies the variability by looking at the range covered by ALL the data, the IQR measures the variability of a distribution by giving us the range covered by the MIDDLE 50% of the data.
- **$IQR = Q3 - Q1$**



EXAMING DISTRIBUTIONS

Here is how the IQR is actually found:

1. Arrange the data in increasing order, and find the median M . Recall that the median divides the data, so that 50% of the data points are below the median, and 50% of the data points are above the median.

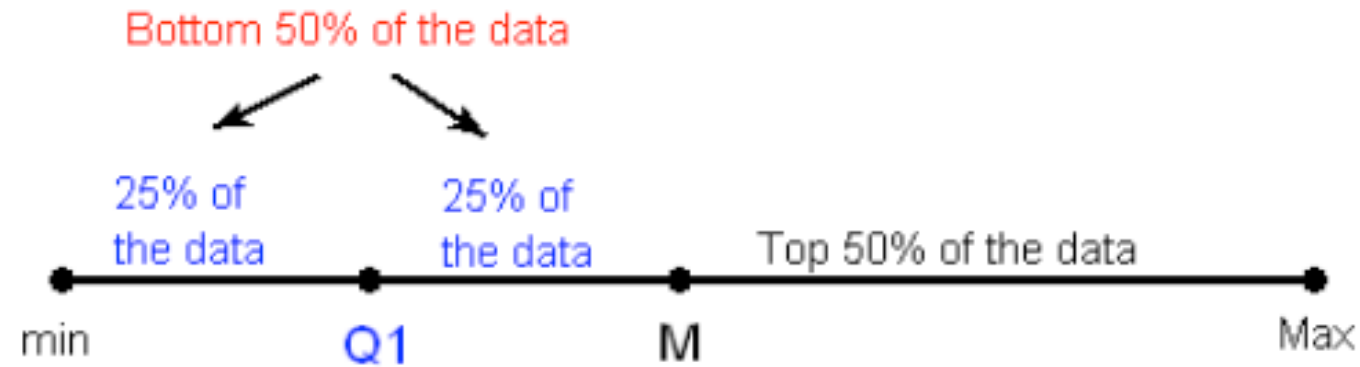


Recall:

Median: is the midpoint of the distribution. It is the number such that half of the observations fall above, and half fall below.

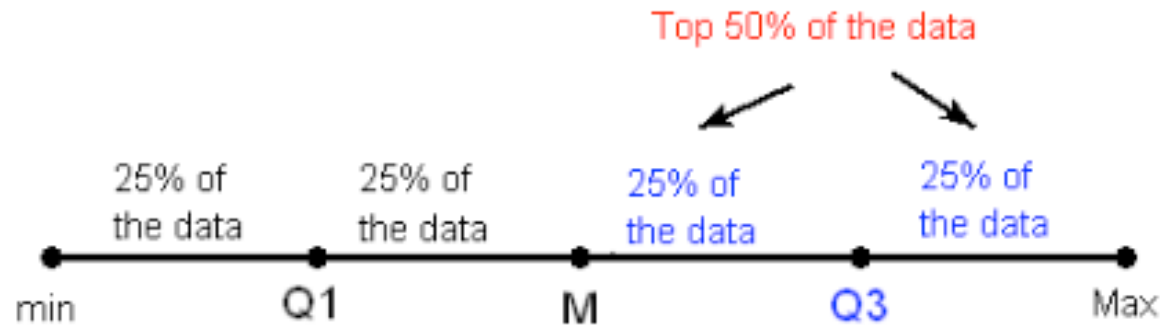
EXAMING DISTRIBUTIONS

2. Find the median of the lower 50% of the data. This is called the first quartile of the distribution, and the point is denoted by Q1. Note from the picture that Q1 divides the lower 50% of the data into two halves, containing 25% of the data points in each half. Q1 is called the first quartile, since one quarter of the data points fall below it.



EXAMING DISTRIBUTIONS

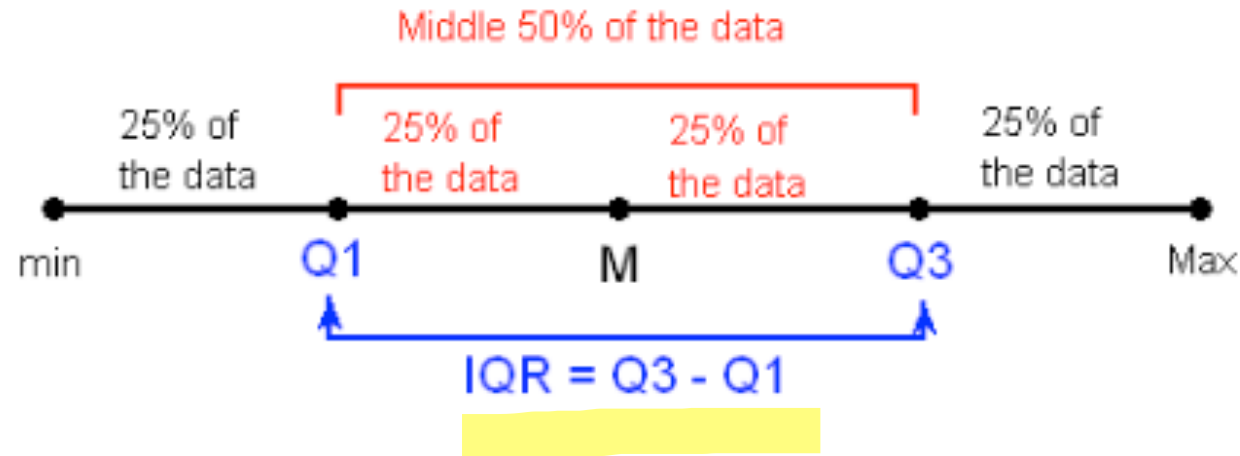
3. Repeat this again for the top 50% of the data. Find the median of the top 50% of the data. This point is called the third quartile of the distribution, and is denoted by Q3. Note from the picture that Q3 divides the top 50% of the data into two halves, with 25% of the data points in each. Q3 is called the third quartile, since three quarters of the data points fall below it.



EXAMING DISTRIBUTIONS

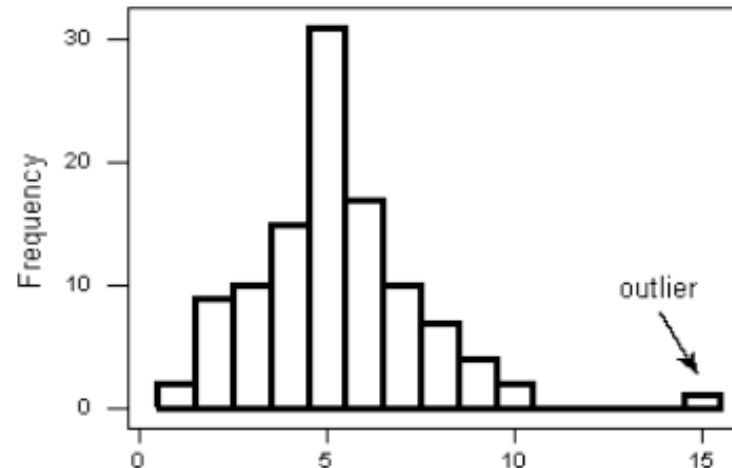
4. The middle 50% of the data falls between Q1 and Q3, and therefore:

$$\text{IQR} = Q3 - Q1$$



EXAMING DISTRIBUTIONS

- The IQR is used as the basis for a rule of thumb for identifying **outliers**.
- Recall:
 - Outliers are observations that fall outside the overall pattern. For example, the following histogram represents a distribution that has a high probable outlier:

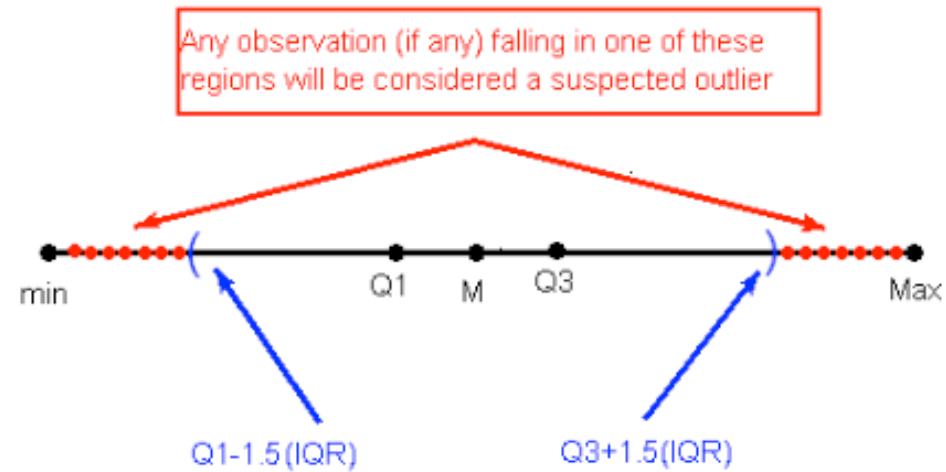


EXAMING DISTRIBUTIONS

The 1.5(IQR) Criterion for Outliers An observation is considered a suspected outlier if it is:

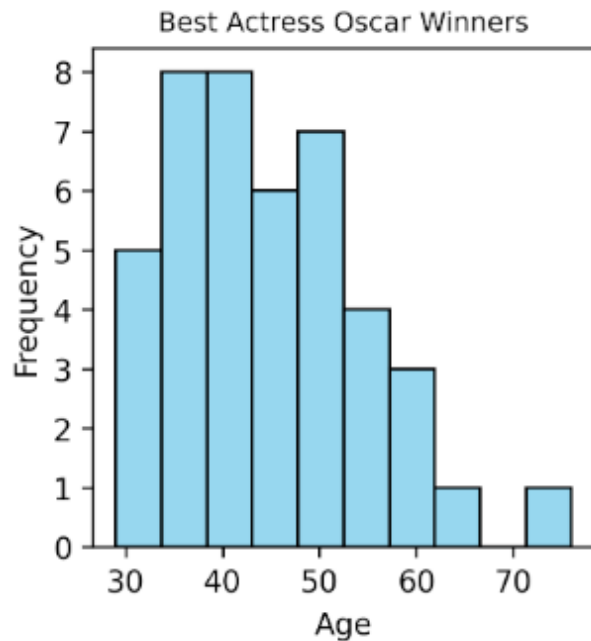
- below $Q1 - 1.5(IQR)$ or
- above $Q3 + 1.5(IQR)$

The following picture illustrates this rule:



EXAMING DISTRIBUTIONS

- Example (C) Best Actress Oscar Winners
- 34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45
49 39 34 26 25 33 35 35 28 30 29 61 32 33 45 29 62 22 44



- For this example we found that $Q1 = 30.5$ and $Q3 = 42.5 \Rightarrow IQR = 11.5$.
- $Q1 - 1.5(IQR) = 30.5 - (1.5)(11.5) = 13.25$.
- $Q3 + 1.5(IQR) = 42.5 + (1.5)(11.5) = 59.25$.

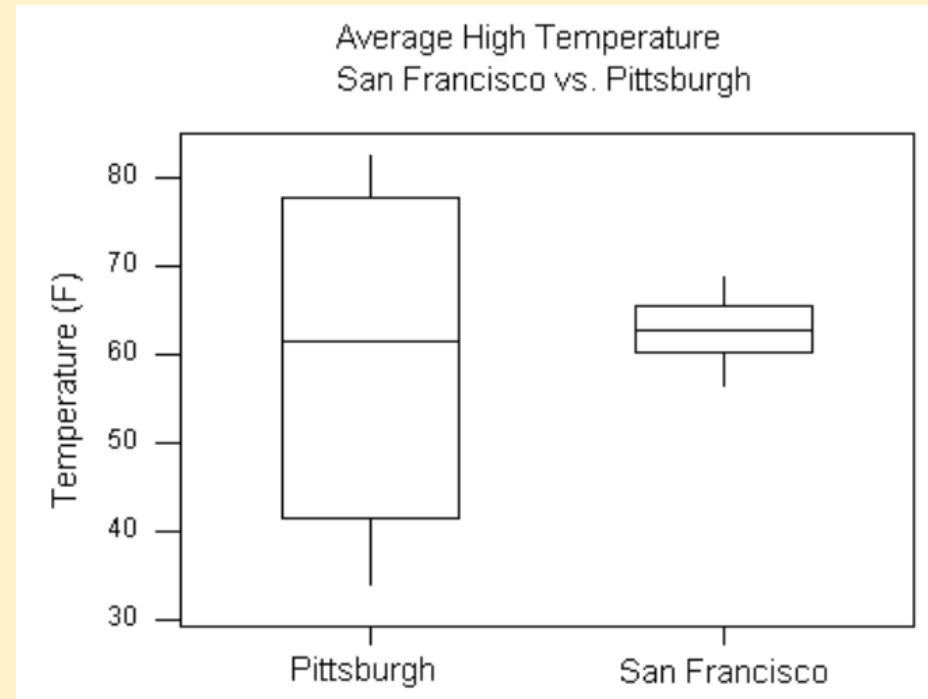
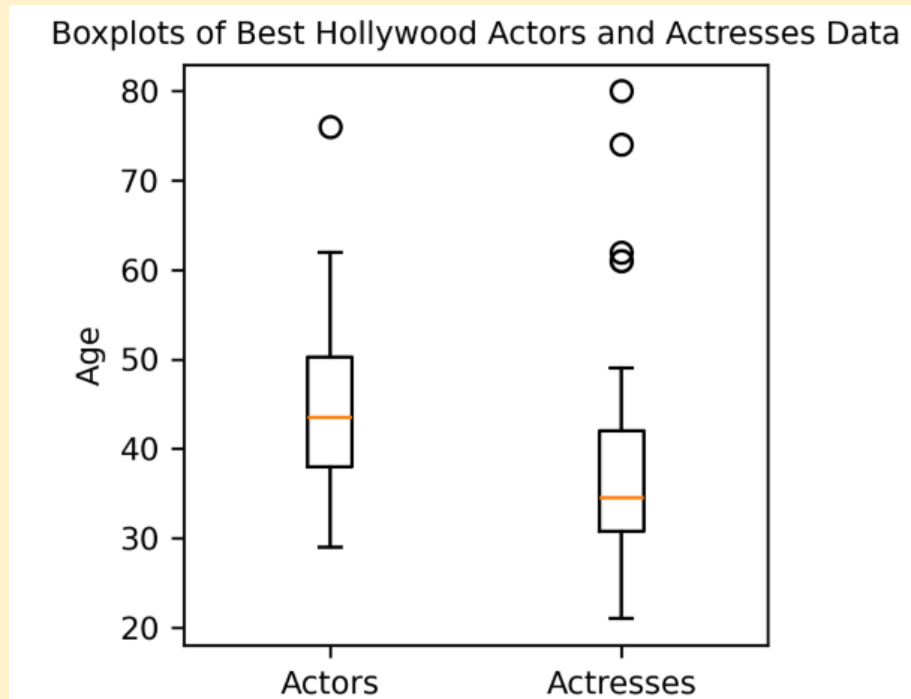
The $1.5(IQR)$ criterion tells us that any observation that is below 13.25 or above 59.25 is considered a suspected outlier.

Therefore, the observations 61, 61, 62, 74 and 80 should be flagged as suspected outliers in the distribution of ages.

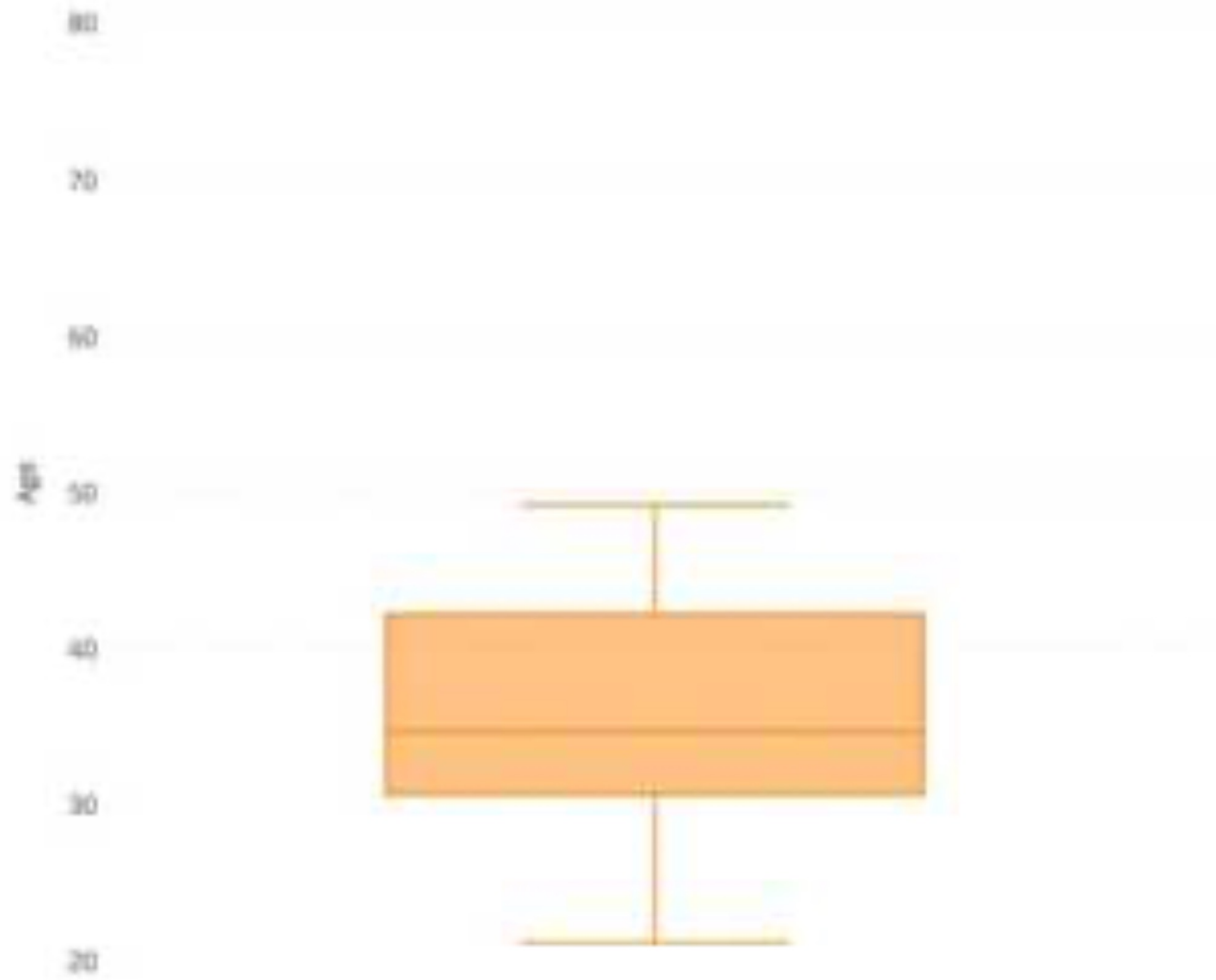
Exploratory Data Analysis

One Quantitative Variable

2. **Boxplot** - graphically represents the distribution of a quantitative variable by visually displaying the five-number summary* and any observation that was classified as a suspected outlier using the $1.5(\text{IQR})$ criterion.



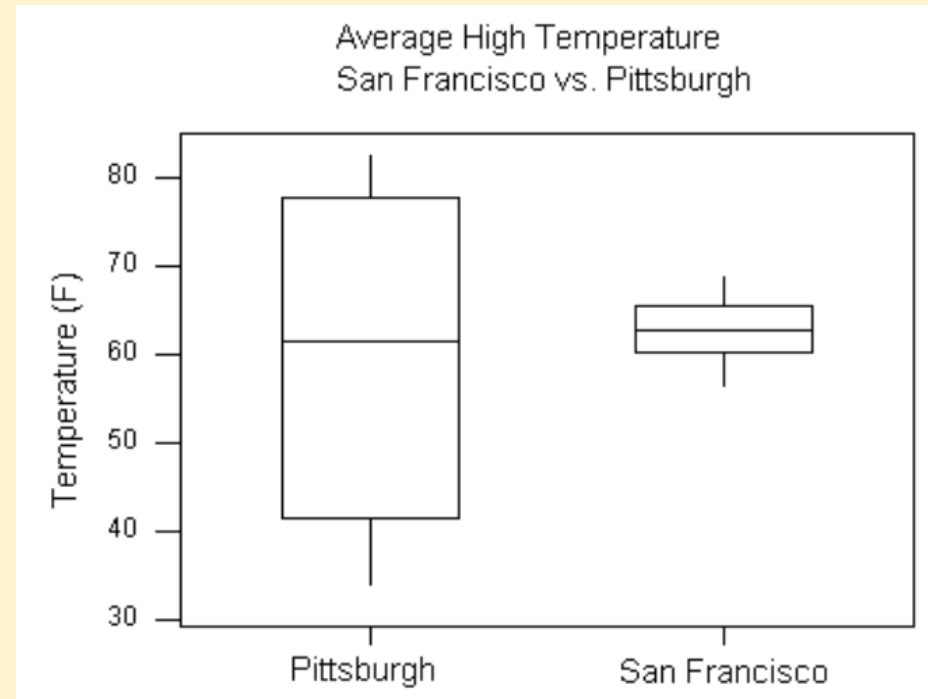
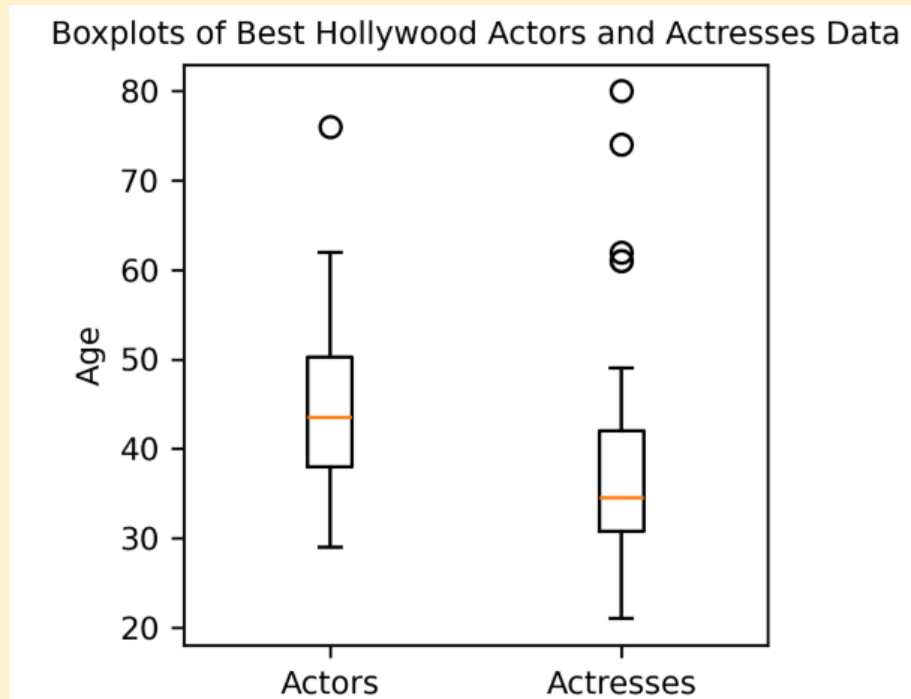
*The five number summary: the extremes (min & Max), which provide the range covered by all the data; and the quartiles (Q1, M, and Q3), which together provide the IQR, the range covered by the middle 50% of the data.



Exploratory Data Analysis

One Quantitative Variable

2. **Boxplot** - graphically represents the distribution of a quantitative variable by visually displaying the five-number summary* and any observation that was classified as a suspected outlier using the $1.5(\text{IQR})$ criterion.



*The five number summary: the extremes (min & Max), which provide the range covered by all the data; and the quartiles (Q1, M, and Q3), which together provide the IQR, the range covered by the middle 50% of the data.

Exploratory Data Analysis

- **One Quantitative Variable**

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**

Example: Video Store Customers

The following numbers of customers who entered a video store in 8 consecutive hours: 7, 9, 5, 13, 3, 11, 15, 9

1. Find the mean, $\bar{x}$ of your data: $\frac{7 + 9 + 5 + \dots + 9}{8} = 9$
2. Find the deviations from the mean: the difference between each observation and the mean
(7 – 9), (9 – 9), (5 – 9), (13 – 9), (3 – 9), (11 – 9), (15 – 9), (9 – 9)
– 2, 0, – 4, 4, – 6, 2, 6, 0

Exploratory Data Analysis

- One Quantitative Variable

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**

Example: Video Store Customers

The following numbers of customers who entered a video store in 8 consecutive hours: 7, 9, 5, 13, 3, 11, 15, 9

3. Square each of the deviations:

The first few are

$(-2)^2 = 4$, $(0)^2 = 0$, $(-4)^2 = 16$, and the rest are 16, 36, 4, 36, 0.

4. Average the square deviations by adding them up, and dividing by $n - 1$, (one less than the sample size):

$$\frac{4 + 0 + 16 + 36 + 4 + 36 + 0}{8 - 1} = \frac{112}{7} = 16.$$

Exploratory Data Analysis

- One Quantitative Variable

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**

Example: Video Store Customers

The following numbers of customers who entered a video store in 8 consecutive hours: 7, 9, 5, 13, 3, 11, 15, 9

5. The SD of the data is the square root of the variance: $SD = \sqrt{16} = 4$.

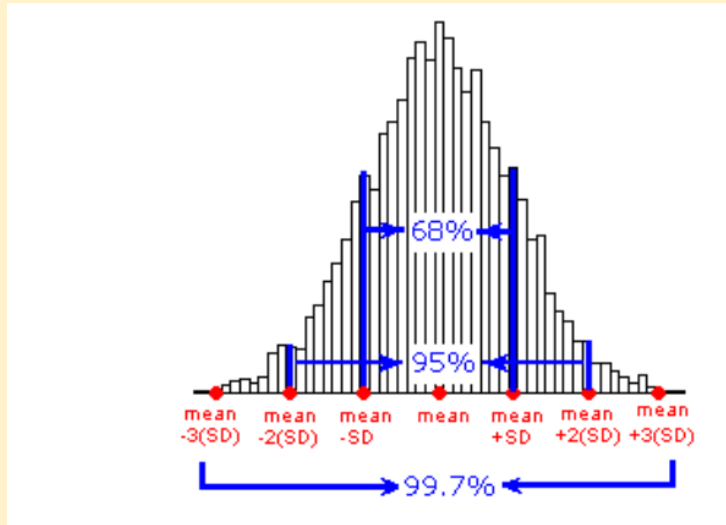
- Why do we take the square root? Note that 16 is an average of the squared deviations, and therefore has different units of measurement. In this case 16 is measured in “squared customers,” which obviously cannot be interpreted. We therefore take the square root in order to compensate for the fact that we squared our deviations, and in order to go back to the original unit of measurement.

Recall that the average number of customers who enter the store in an hour is 9. The interpretation of $SD = 4$ is that on average, the actual number of customers that enter the store each hour is 4 away from 9.

Exploratory Data Analysis

One Quantitative Variable

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**

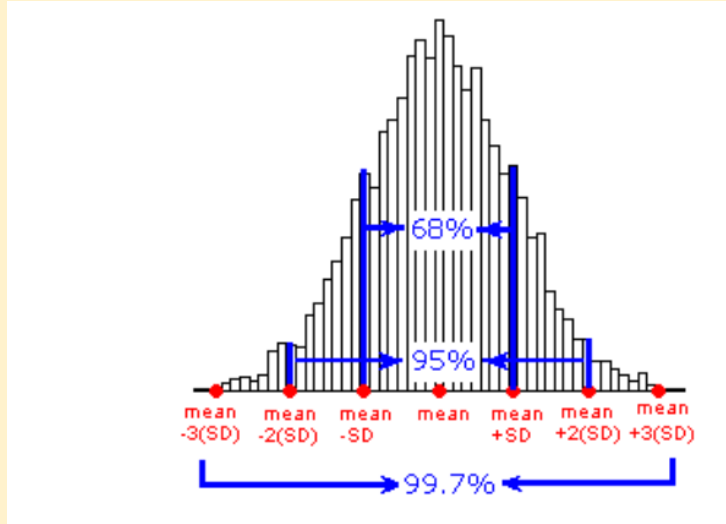


- The standard deviation measures the spread by reporting a typical (average) distance between the data points and their average.
- It is appropriate to use the SD as a measure of spread with the mean as the measure of center.
- Recall:
 - Mean:** is the average of a set of observations (i.e., the sum of the observations divided by the number of observations).

Exploratory Data Analysis

- One Quantitative Variable

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**

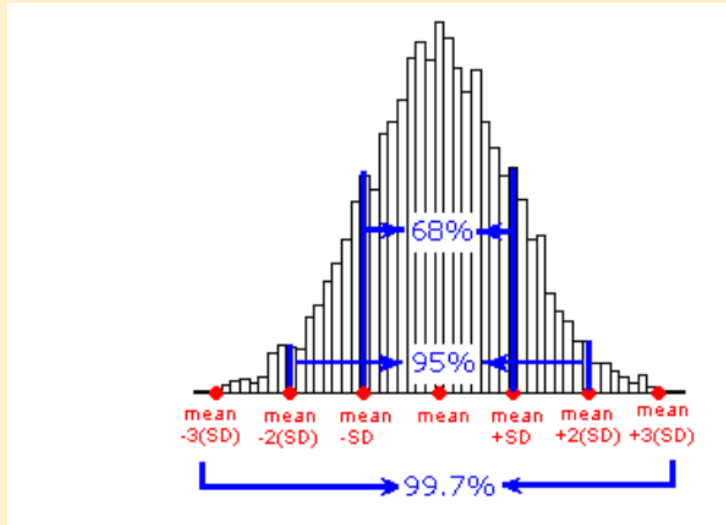


- Since the mean and standard deviations are highly influenced by extreme observations, they should be used as numerical descriptions of the center and spread only for distributions that are roughly symmetric.

Exploratory Data Analysis

- One Quantitative Variable

- 3 most commonly used measures of spread – 1) Range, 2) Inter-quartile range (IQR) & 3) **Standard deviation**



- For symmetric mound-shaped distributions, the Standard Deviation Rule tells us what percentage of the observations falls within 1, 2, and 3 standard deviations of the mean, and thus provides another way to interpret the standard deviation's value for distributions of this type.

EXERCISES

 1. A survey taken in a large statistics class contained the question: “What’s the fastest you have driven a car (in miles per hour)?” The five-number summary for the 87 males surveyed is:

min = 55, Q1 = 95, Median = 110, Q3 = 120, Max = 155

Should the largest observation in this data set be classified as an outlier?

- A. Yes.
- B. No.

1. A survey taken in a large statistics class contained the question: “What’s the fastest you have driven a car (in miles per hour)?” The five-number summary for the 87 males surveyed is:

min = 55, Q1 = 95, Median = 110, Q3 = 120, Max = 155

Should the largest observation in this data set be classified as an outlier?

A. Yes.

B. No.

- The IQR in this case is $120 - 95 = 25$. Applying the $1.5(\text{IQR})$ rule, we find: $Q3 + 1.5(\text{IQR}) = 120 + 1.5(25) = 157.5$, and therefore the largest observation, 155, **should not be classified** as an outlier.
- Note, however that in this case: $Q1 - 1.5(\text{IQR}) = 95 - 1.5(25) = 57.5$, and therefore the smallest observation, 55, should be classified as an outlier.

2. A survey taken of 140 sports fans asked the question: "What is the most you have ever spent for a ticket to a sporting event?"

The five-number summary for the data collected is:

min = 85, Q1 = 130, Median = 145, Q3 = 150, Max = 250

Should the smallest observation be classified as an outlier?

- A. Yes.
- B. No.

2. A survey taken of 140 sports fans asked the question: "What is the most you have ever spent for a ticket to a sporting event?"

The five-number summary for the data collected is:

min = 85, Q1 = 130, Median = 145, Q3 = 150, Max = 250

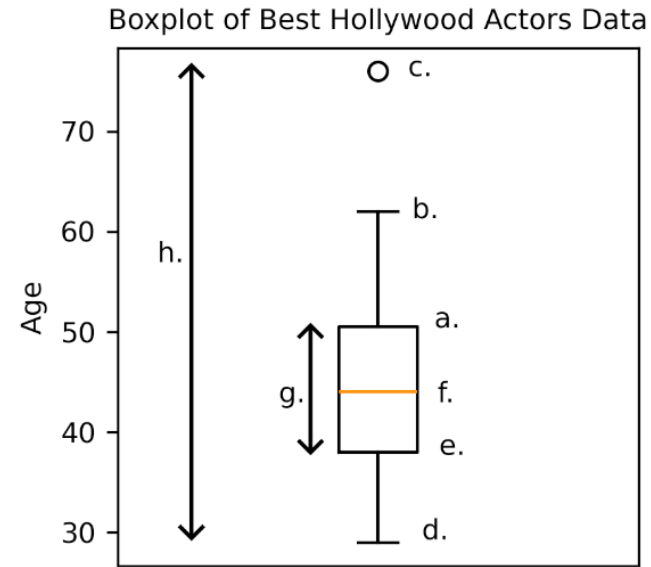
Should the smallest observation be classified as an outlier?

A. Yes.

B. No.

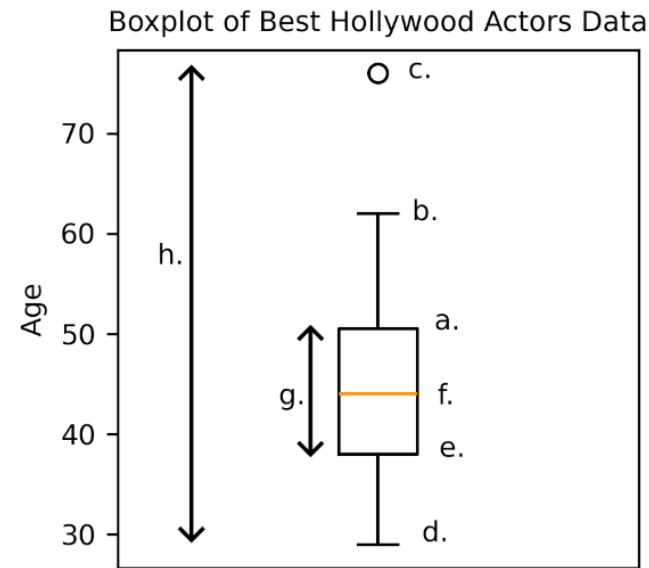
A. Correct. The IQR is $150 - 130 = 20$. Using the $1.5(\text{IQR})$ criterion we get $130 - 1.5(20) = 100$. Since the smallest observation of 85 is smaller than 100, it should be considered an outlier.

3. Match the points on the boxplot represented by the letters going from a to h to the list below:



- A. Q1.
- B. Q3.
- C. M.
- D. Min.
- E. Max.
- F. IQR.
- G. Range.
- H. None of the above.

3. Match the points on the boxplot represented by the letters going from a to h to the list below:



- A. Q1.
- B. Q3.
- C. M.
- D. Min.
- E. Max.
- F. IQR.
- G. Range.
- H. None of the above.

a. Q3. This is the third quartile. 75% of the observations are smaller than this number.

b. None of the above. This is the largest observation that is not an outlier.

c. Max. This is the Maximum and in this example it includes an outlier.

d. Min. This is the Minimum.

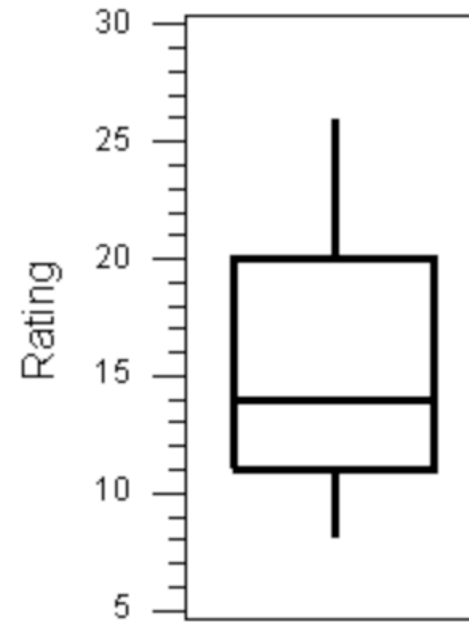
e. Q1. This is the first quartile.

f. M. This is the median.

g. The range is the distance between the Min and the Max.

h. This is the interquartile range (IQR)

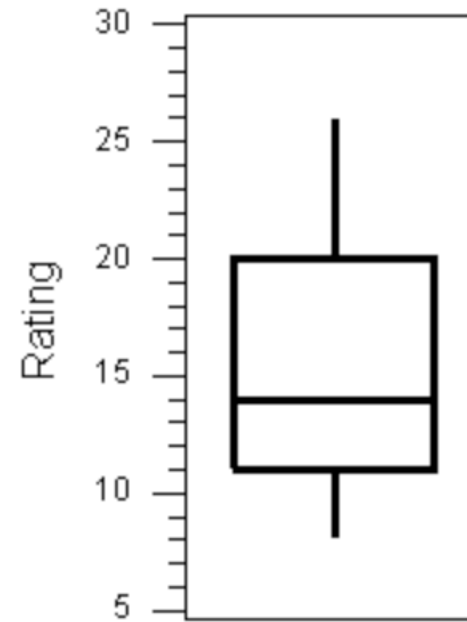
4. The boxplot below displays ratings for TV shows during sweeps week:



50% of the shows have a rating greater than which value?

- A. 11.
- B. 15.5.
- C. 14.
- D. 17.

 4. The boxplot below displays ratings for TV shows during sweeps week:



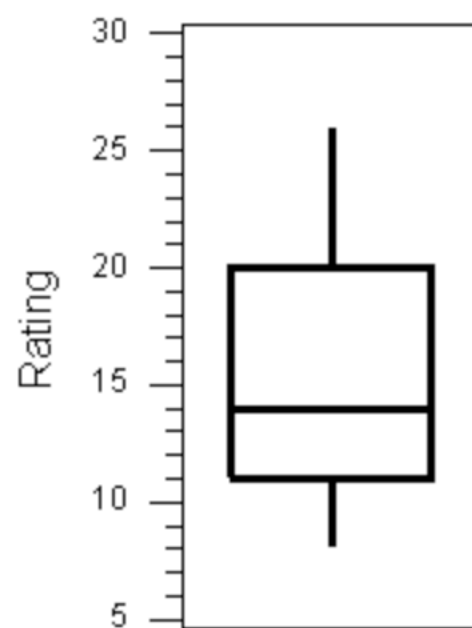
50% of the shows have a rating greater than the median (M), which, according to the boxplot, is 14.

50% of the shows have a rating greater than which value?

- A. 11.
- B. 15.5.
- C. 14.**
- D. 17.

5. What is the percentage of shows that have a rating of less than 20?

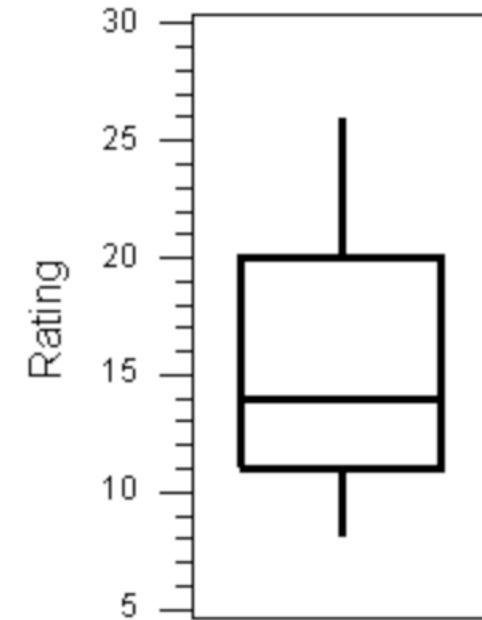
- A. 67%.
- B. 75%.
- C. 84%.
- D. 25%.



5. What is the percentage of shows that have a rating of less than 20?

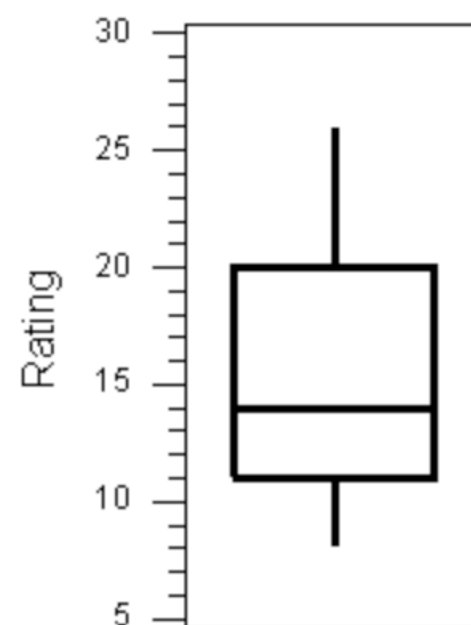
- A. 67%.
- B. 75%.
- C. 84%.
- D. 25%.

The boxplot indicates that 20 is the third quartile (Q3) of the rating distribution, and therefore, by definition, three quarters of the observations, or 75%, of the shows have a rating below it.



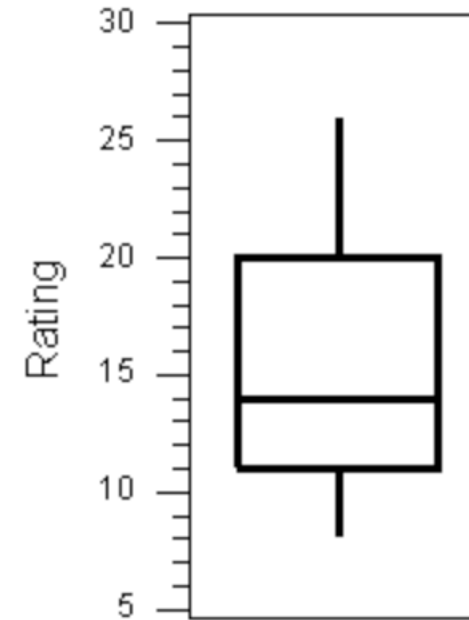
6. Within which interval of rating would you expect to find the **largest** number of shows?

- A. 5-10.
- B. 10-15.
- C. 15-20.
- D. 20-25.
- E. All are equal.



6. Within which interval of rating would you expect to find the **largest** number of shows?

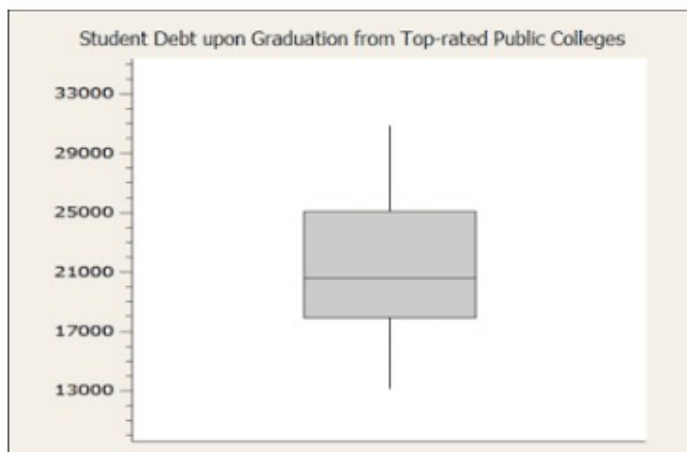
- A. 5-10.
- B. 10-15.
- C. 15-20.
- D. 20-25.
- E. All are equal.



Note that 10 is below Q1, and 15 is above the M, so we expect more than 25% of the shows to have ratings in that interval.

Using a similar reasoning, in all the other intervals we would expect to see LESS than 25% of the shows.

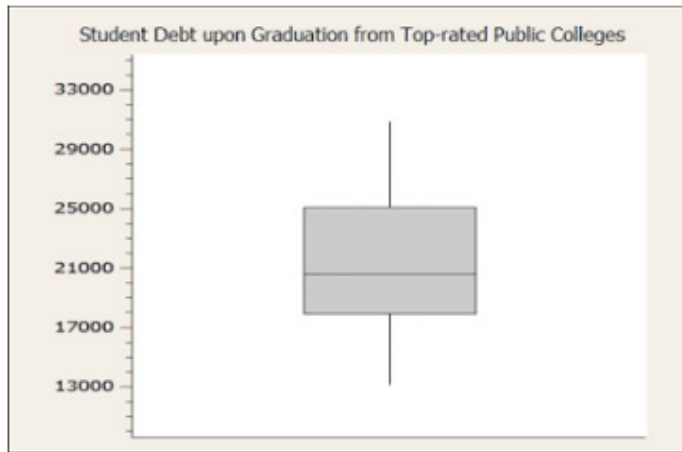
The Web site www.kiplinger.com published the “Best Values in Public Colleges,” saying they were determined by rankings based on data from more than 500 public four-year colleges and universities. Using out-of-state rankings that considered the academic quality, total out-of-state costs, and average costs after financial aid, the 50 best-rated colleges were selected. The boxplot pictured displays the average debt per graduate for these colleges.



7. What percentage of the graduates will have a debt greater than \$25,000?

- A. 25%.
- B. 50%.
- C. 75%.
- D. 81%.
- E. 19%.

The Web site www.kiplinger.com published the “Best Values in Public Colleges,” saying they were determined by rankings based on data from more than 500 public four-year colleges and universities. Using out-of-state rankings that considered the academic quality, total out-of-state costs, and average costs after financial aid, the 50 best-rated colleges were selected. The boxplot pictured displays the average debt per graduate for these colleges.



A. Since \$25,000 is the third quartile (Q3), one-quarter of the graduates will have a debt greater than this amount.

7. What percentage of the graduates will have a debt greater than \$25,000?

A. 25%.

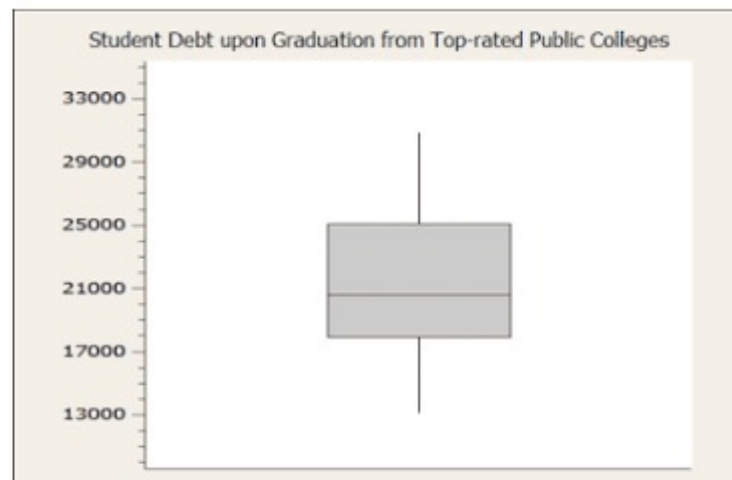
B. 50%.

C. 75%.

D. 81%.

E. 19%.

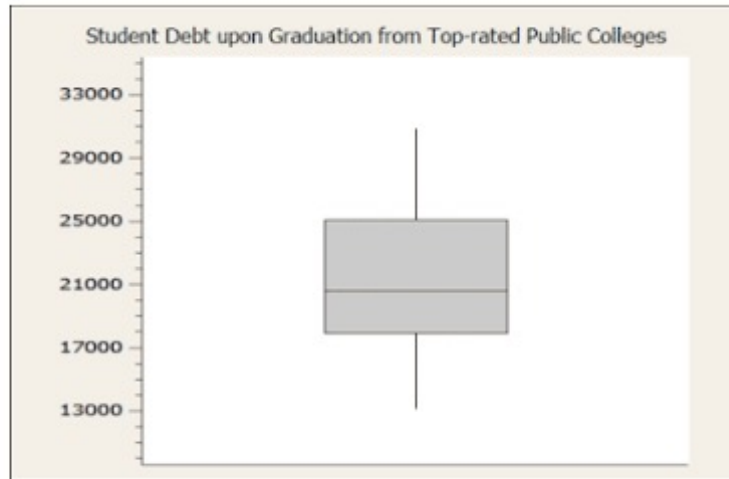
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8. 50% of the debts owed are smaller than what amount?

- A. \$18,000.
- B. \$20,600.
- C. \$22,000.
- D. \$25,000.

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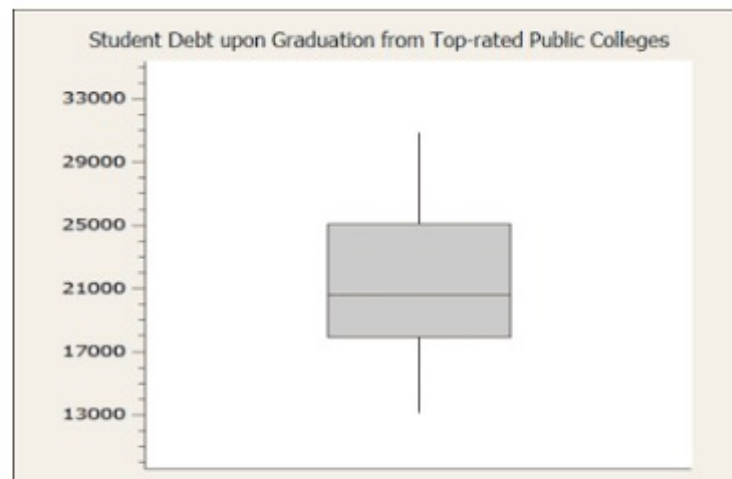


B. 50% of the debts are smaller than the median, which according to the boxplot is between 20,000 and 21,000.

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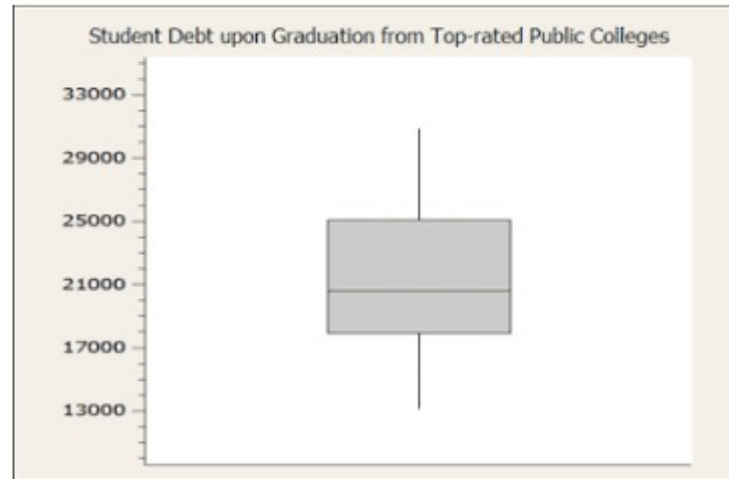
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9. Within which interval of debts would you expect to find the **largest** number of graduates?

- A. 13,000–17,000.
- B. 17,000–21,000.
- C. 21,000–25,000.
- D. 25,000–29,000.
- E. All the intervals are equal.

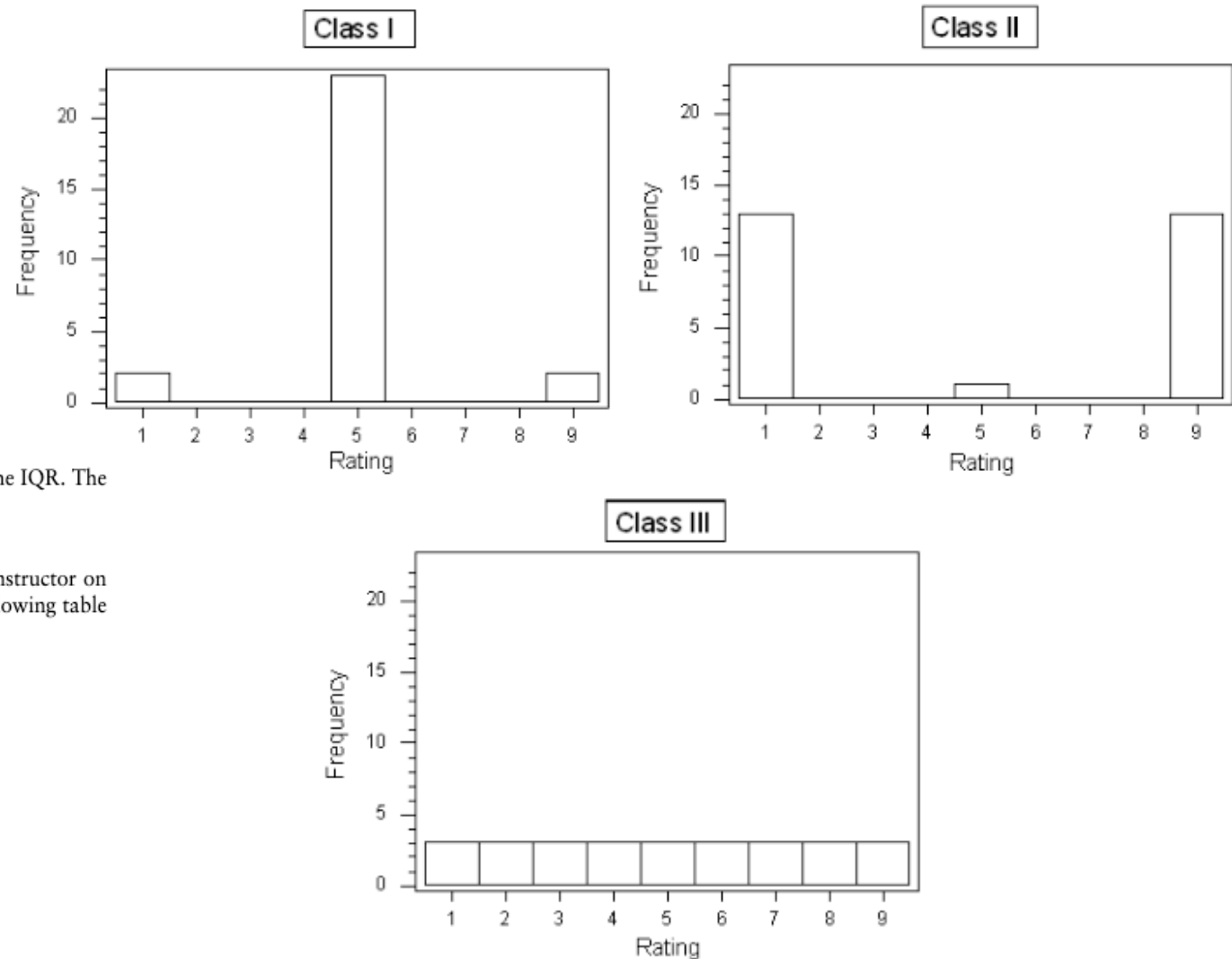
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B. Since 17,000 is below the first quartile and 21,000 is above the median, more than 25% of the debts are in this interval.

9. Within which interval of debts would you expect to find the **largest** number of graduates?

- A. 13,000–17,000.
- B. 17,000–21,000.**
- C. 21,000–25,000.
- D. 25,000–29,000.
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10. The concept of standard deviation is less intuitive as a measure of spread than the range or the IQR. The following activity is designed to help you develop a better intuition for the standard deviation.

Background At the end of a statistics course, the 27 students in the class were asked to rate the instructor on a number scale of 1 to 9 (1 being “very poor,” and 9 being “best instructor I’ve ever had”). The following table provides three hypothetical rating data:

Rating	1	2	3	4	5	6	7	8	9
<i>Class I</i>	1	0	0	0	22	0	0	0	1
<i>Class II</i>	12	0	0	0	1	0	0	0	12
<i>Class III</i>	2	2	2	2	2	2	2	2	2

And here are the histograms of the data:

Assume that the average rating in each of the three classes is 5 (which should be visually reasonably clear from the histograms), and recall the interpretation of the SD as a “typical” or “average” distance between the data points and their mean. Judging from the table and the histograms, which class would have the **largest** standard deviation, and which one would have the **smallest** standard deviation? Explain your reasoning.

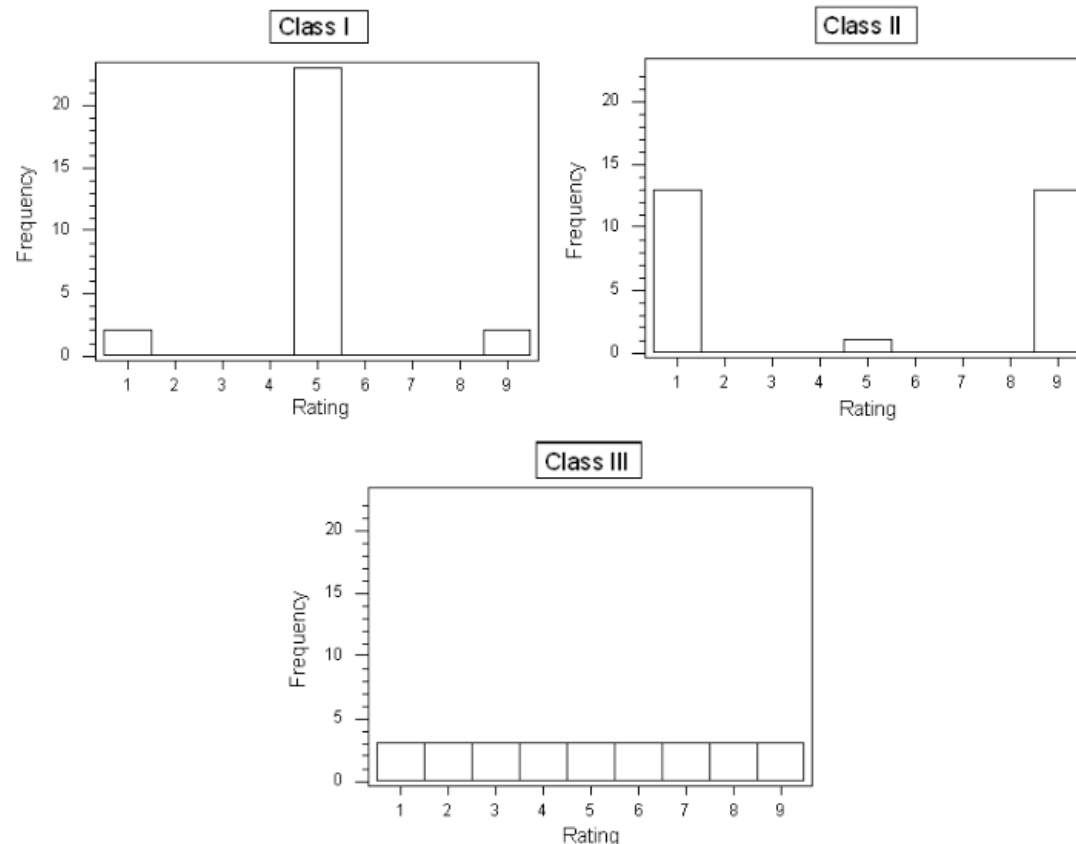
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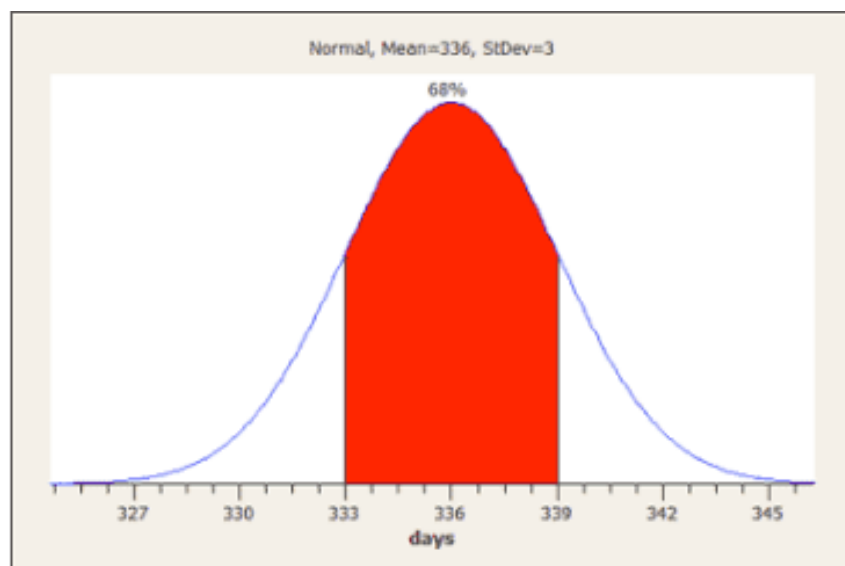
In class I, almost all the ratings are 5, which is also the mean. The average distance between the observations and the mean, then, would be very small. In class II most of the observations are far from the mean (at 1 or 9). The average distance between the observations and the mean in this case would be larger. Class III is the case where some of the observations are close to the mean, and some are far, so the average distance between the observations and the mean would be somewhere in between class I and II. This observation would lead me to conclude that the standard deviation would be ranked (from smallest to largest): Class I, Class III, Class II.



Assume that the average rating in each of the three classes is 5 (which should be visually reasonably clear from the histograms), and recall the interpretation of the SD as a “typical” or “average” distance between the data points and their mean. Judging from the table and the histograms, which class would have the **largest** standard deviation, and which one would have the **smallest** standard deviation? Explain your reasoning.

In general, the larger the animal, the longer the length of pregnancy (also called *gestation period*). For the horse, for example, the gestation period varies roughly according to a normal distribution with a mean of 336 days and a standard deviation of 3 days (Source: These figures are from Moore and McCabe, *Introduction to the Practice of Statistics*).

Use the Standard Deviation Rule to answer the following questions. This picture of the SD rule applied to this distribution will help:

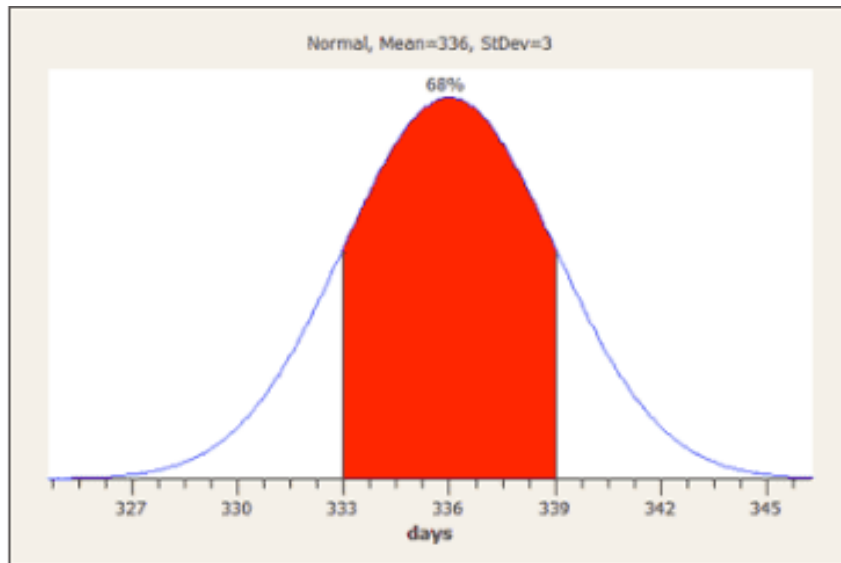


11. Almost **all (99.7%)** horse pregnancies fall in what range of lengths?

- A. Above 336 days.
- B. Below 336 days.
- C. Between 333 and 339 days.
- D. Between 330 and 342 days.
- E. Between 327 and 345 days.

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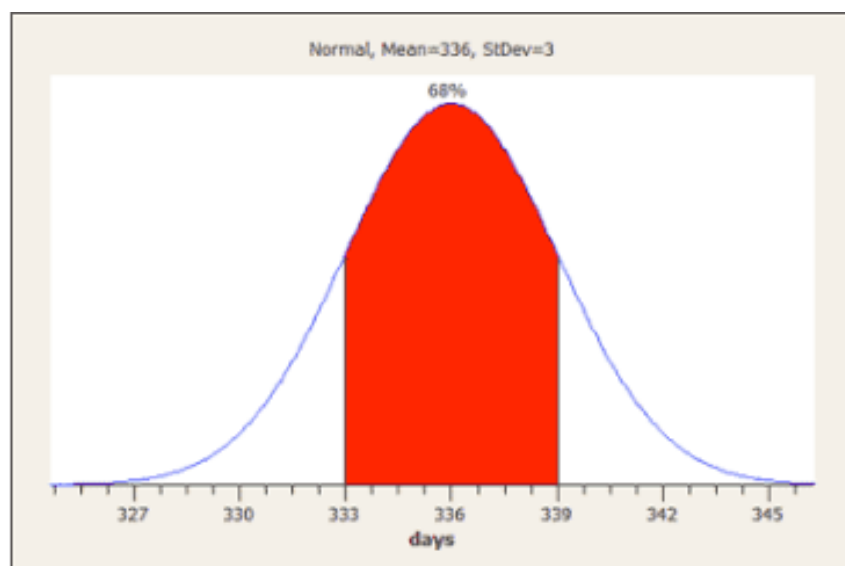
11. Almost **all (99.7%)** horse pregnancies fall in what range of lengths?

- A. Above 336 days.
- B. Below 336 days.
- C. Between 333 and 339 days.
- D. Between 330 and 342 days.
- E. Between 327 and 345 days.**

The Standard Deviation Rule tells us that virtually all the data fall within 3 standard deviations of the mean, which in this case is exactly between $336 - 3(3) = 327$, and $336 + 3(3) = 345$.

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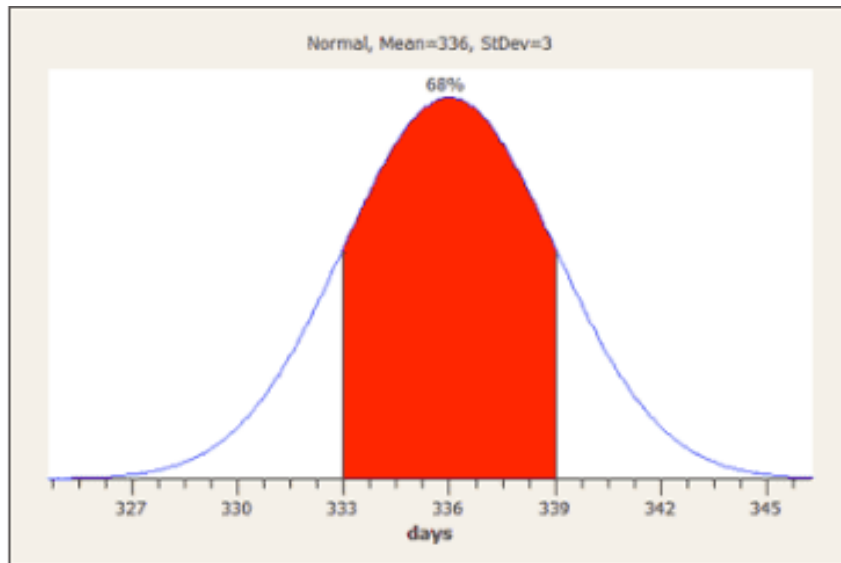


12. What percentage of horse pregnancies last longer than 339 days?

- A. 2.5%.
- B. 16%.
- C. 50%.
- D. 68%.
- E. 100%.

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Use the Standard Deviation Rule to answer the following questions. This picture of the SD rule applied to this distribution will help:



12. What percentage of horse pregnancies last longer than 339 days?

- A. 2.5%.
- B. 16%.
- C. 50%.
- D. 68%.
- E. 100%.

According to the SD rule, 68% of horse pregnancies last between $336 - 3 = 333$ and $336 + 3 = 339$ days, which means that the remaining 32% of horse pregnancies are divided evenly between lasting less than 333 days and lasting more than 339 days. We therefore conclude that 16% of horse pregnancies last more than 339 days.

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 13. About **95%** of the temperatures of dogs fall within which range?

- A. $99.5\text{--}103.1$.
- B. $100.1\text{--}101.9$.
- C. $100.1\text{--}102.5$.
- D. $100.7\text{--}101.9$.
- E. Above 101.3 .

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 13. About **95%** of the temperatures of dogs fall within which range?

- A. 99.5–103.1.
- B. 100.1–101.9.
- C. 100.1–102.5.**
- D. 100.7–101.9.
- E. Above 101.3.

The Standard Deviation Rule tells us that 95% of the temperatures fall within 2 SD of the mean. In this case, this is between $101.3 - 2(0.6) = 100.1$ and $101.3 + 2(0.6) = 102.5$

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 14. What percentage of the temperatures are less than 99.5°F ?

- A. 0.15%.
- B. 0.3%.
- C. 2.5%.
- D. 5%.
- E. 16%.

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 14. What percentage of the temperatures are less than 99.5°F ?

- A. 0.15%.
- B. 0.3%.**
- C. 2.5%.
- D. 5%.
- E. 16%.

According to the SD rule, 99.7% of dog temperatures are between $101.3 - 3(0.6) = 99.5$ and $101.3 + 3(0.6) = 103.1^{\circ}\text{F}$, which means that the remaining 0.3% of temperatures are divided evenly between less than 99.5°F and more than 103.1°F . We therefore conclude that 0.15% of temperatures are less than 99.5°F .

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 15. What percentage of the temperatures are more than 100.1°F ?

- A. 0.15%.
- B. 2.5%.
- C. 84%.
- D. 95%.
- E. 97.5%.

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F . Use this information to answer the following questions:

 15. What percentage of the temperatures are more than 100.1°F ?

- A. 0.15%.
- B. 2.5%.
- C. 84%.
- D. 95%.
- E. 97.5%.

Since 100.1°F is 2 SD below the mean, the percentage between 2 SD below the mean and 2 SD above the mean is 95%. This means that 95% of the temperatures are between $101.3 - 2(0.6) = 100.1$ and $101.3 + 2(0.6) = 102.5$. The remaining 5% is divided evenly between below 100.1 and above 102.5°F . Therefore, the percentage that is more than 100.1 is $95\% + 2.5\% = 97.5\%$.